



Quantum Ledger Technology (QLT)

A Quantum-Secure Anchor for DeFi

Helping the DeFi industry save billions in yearly losses



AWARDED SEAL OF EXCELLENCE BY THE EUROPEAN COMMISSION

The Crypto Economy: Exploding Growth, Soaring Stakes

Global Market Value Surpasses \$3 Trillion

The blockchain-powered crypto economy has already crossed the €3 trillion mark, establishing itself as a major force in global finance.

Over 1 Billion Users, 10,000+ Active Cryptos

Over 1 billion users trade 10k+ tokens—a rapidly growing ecosystem of digital assets powering DeFi, NFTs, and Web3 innovation.

Government & Institutional Adoption Rising

Governments and major financial players like BlackRock and Fidelity are entering the crypto space, signaling mainstream acceptance.

Over \$20Bn Lost Cumulatively in Crypto Hacks

Crypto theft jumped 40% last year, with over \$2.3 billion in losses. Security breaches continue to threaten ecosystem stability.

📌 **Critical Insight:** Crypto is scaling fast—and secure, quantum-resistant infrastructure is now the need of the hour to protect this explosive growth.

Why Current Blockchain Infrastructure Remains Insecure

Almost all of the blockchain/crypto hacks can be attributed to one of the following threat :



Client-End Vulnerabilities

Assets lost due to compromised user devices, malware infections, and insecure client-side systems that expose private keys.

Exchange-End Vulnerabilities

Assets lost while using centralized (CEX) and decentralized (DEX) exchanges for crypto transactions due to platform security weaknesses.

Impending Threat from Quantum Computers

Existential risk that quantum computers will break the current Public Key Infrastructure (PKI) ecosystem of blockchain and distributed ledger technology.



Scams and hacks prove detrimental to the growth of the blockchain and crypto economy. Cybersecurity remains the top priority for industry survival and institutional trust.



Two Reasons for All Computer Vulnerabilities

All security breaches in legacy systems—including distributed ledger technology—can be summarized into two major categories that create systemic weaknesses:

User-Facing PKI Cryptographic Access Authentication



All hacks (adversarial or insider), including quantum decryption, happen via PKI-powered access authentication, which is **always user-facing**

Hacks Due to Third-Party Permissions (TPPs)



Inherent vulnerabilities originate from **mandatory TPPs** granted by legacy hardware and software to third-party codes and applications



Unfortunately, current cybersecurity is inherently vulnerable to both attack vectors and is prone to complete breakdown. The industry needs a fundamental architectural shift.

The State of the Art: Post-Quantum Cryptography (PQC)

Only Current Hope: The industry's sole recognized path for futuristic blockchain security.

High Failure Rate: 97% of NIST candidates (80 out of 82) already rejected or withdrawn.

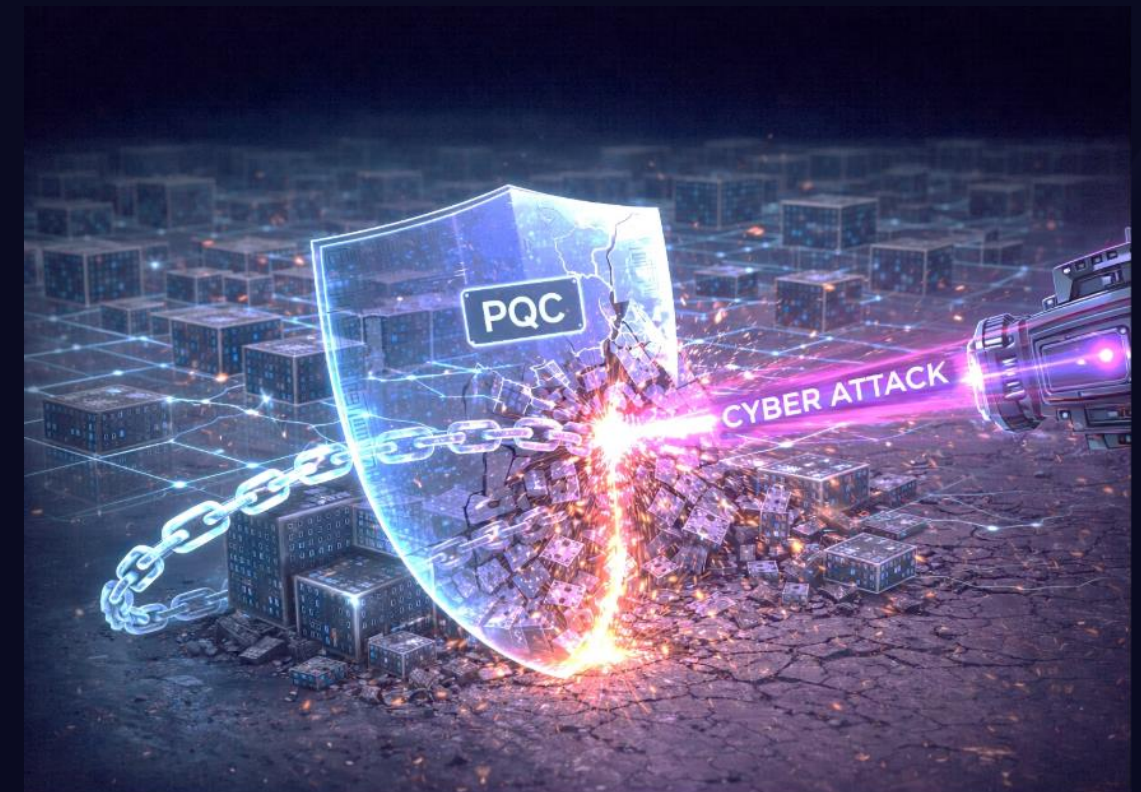
Hardware Gap: Incompatible with 70%–85% of legacy IoT devices.

Resource Heavy: Extremely slow and energy-inefficient compared to current standards.

Flawed Foundation: Still relies on Public Key Infrastructure (PKI) with vulnerable user-facing handshakes.

Unresolved Risks: Fails to eliminate "Third-Party Permissions" (TPPs) at the system level.

Partial Fix: Only upgrades encryption math without securing the underlying hardware architecture.



Twofold Problem, Twofold Solution: QLT

QRUECA

Quantum Resistant User Evasive Cryptographic Authentication solves the legacy user-facing PKI authentication vulnerability problem

ZVC

Zero Vulnerability Computing bans all TPPs, merging legacy software layers to build 3SoC (Solid-State Software on a Chip), delivering AZT (Absolute Zero Trust) to eradicate the classical attack surface.



+



QLT Advantages

- DLT agnostic and quantum resilient
- Inaccessible to legacy systems
- Robust, lightweight, and fast
- Cost and energy-efficient
- Future-ready for all DLT use cases

01



✓ Intellectual Property
Six Patents

02

imec

✓ Tested & Validated by
[IMEC LABS, BELGIUM](#)

03



✓ Supported by 50%
EU [Partners](#)

04



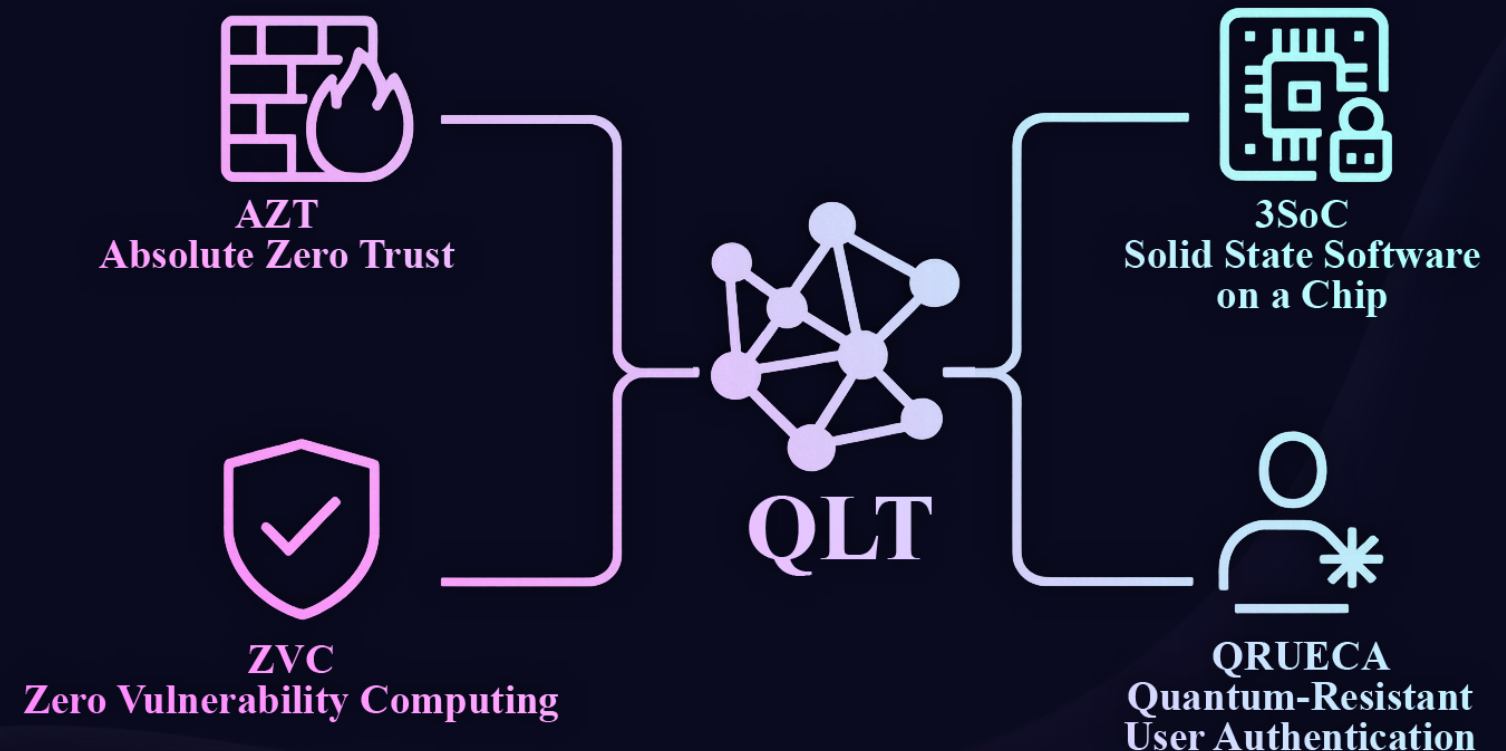
future internet

✓ Published in
[FUTURE INTERNET](#)

QLT Features: Unrivalled Security Beyond Encryption

- 1 ZVC Technology**
Eliminates the majority of malware risks by banning all unauthorized third-party permissions.
- 2 3SoC Architecture**
A tamper-proof "Software-on-a-Chip" that physically hardens security at the hardware level
- 3 QRUECA Authentication**
Replaces vulnerable passwords with "User-Evasive" device-to-device handshakes.
- 4 Quantum-Ready**
Provides lightweight, energy-efficient protection against future quantum computing threats.
- 5 Universal Integration**
A protocol-agnostic anchor that secures any blockchain or exchange without upgrades.

Quantum Ledger Technology Features



Quantum-Resilience of QLT Without PQC



User-Device Handshake



Device-Device Handshake

Legacy Access Authentication

- Always user-facing cryptography
- the first handshake is user-controlled
- exposed and vulnerable to quantum threats.

QLT's QRUECA Access Authentication

- User-evasive cryptography
- first handshake is **Device-Device** using QCC (QRUECA Crypto Certificate),
- second handshake excludes all non-QLT nodes.

❑ QLT is super light, economical, fast, and energy-efficient compared to expensive, slow, and energy-inefficient PQC implementations.

QLT Product Offerings

PHASE 1

The QLT Hardware Wallet: A Revolution in Crypto Security

Timeline: Next 6 Months

Launching QLT-powered **Crypto Hardware Wallet**. (First tests completed at IMEC Labs, Belgium, will integrate Magpie services to allow for seamless asset management and trading of popular tokens like BTC and ERC20.



Subscriber



Client 3SoC Devices

QLT Tunnel



Network Nodes

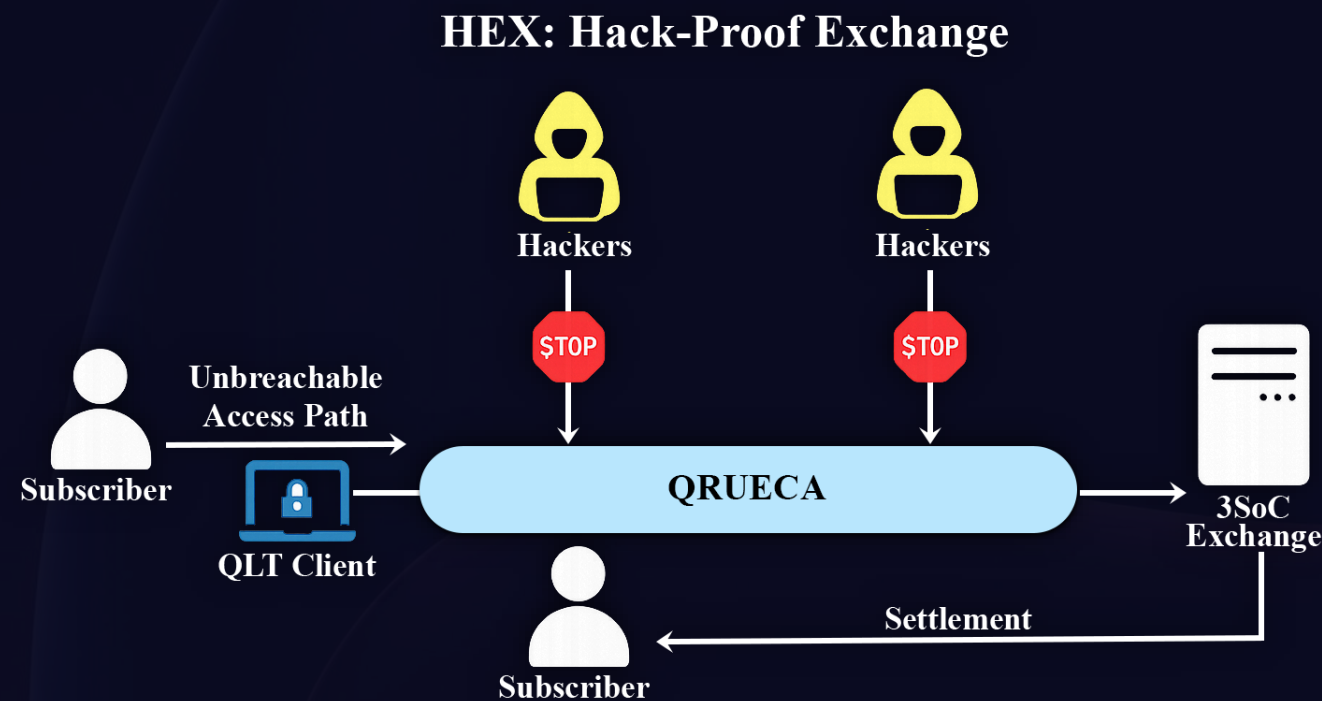
QLT Product Offerings

PHASE 2

QLT Server-End Deployment for Blockchain Nodes and Exchanges

Timeline: Post 6 Months

QLT-powered DLT nodes and Exchange Servers. This will enhance the security of the core blockchain infrastructure and crypto exchanges, thereby strengthening the entire DeFi ecosystem.



QLT : A Convergence of Three Brands

- 1 Liquidity Aggregator raised ~\$6 Million**
Decentralized liquidity-aggregation protocol design for cross-chain and on-chain swaps supporting 23+ Chains
- 2 Vibrant Community of 440K Crypto users**
Aggregator supports users of 23 Chains & Fjord, funded 400 projects
- 3 Fjord Crypto Launchpad Acquisition in Process**
Committed \$4M to acquire Fjord, which raised \$1B for DeFi projects
- 4 Total Transaction Volumes: ~\$10B (\$9B+1.5B)**
Magpie Aggregator volume=\$9B, Fjord Launchpad volume=\$1.5B
- 5 Winner of Seal of Excellence from EU with Six Pending Patents**
QLT is powered by ZVC winner of Seal of Excellence European Commission



Future of DeFi



Fjord Foundry



The Quantum Future

Early Support

Strategic backing from leading blockchain infrastructure providers and industry pioneers validates QLT's quantum-resistant security approach.



AUTONIO FOUNDATION LTD.

Letter of Intent

Autonio Foundation Ltd.
5 High Street, 2nd Floor, Bristol
BS9 3BY, UK

Date: Aug 8, 2021

Subject: Mutual interest expressed vide this letter of intent

Dear Fazal Raheman,

With reference to your email dated 28th July, 2021, we, Autonio Foundation Ltd., UK, are interested in placing an initial order of 10,000 NatiVault (NV) devices subject to the following terms and conditions:

1. Blockchain 5.0 satisfies the functional requirements stated in your product specifications.
2. The ZVC patent-pending technology has been validated at Europe's research centre, IMEC.

Sincerely,

Ikram Ansari
Ikram ansari
Head of products
Autonio foundation

autonio.foundation

The Bristol Office, 2nd Floor 5 High Street, Westbury On Trym, Bristol, United Kingdom, BS9 3BY

ΠΑΝΕΠΙΣΤΗΜΙΟ ΠΕΙΡΑΙΩΣ
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Director of SecLab (<https://seclab.cs.unipi.gr/>)
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Email: pkotzani@unipi.gr
Web : <http://www.cs.unipi.gr/pkotzani>

30 September 2021

Letter of support to "ZVC (Zero Vulnerability Computing): A New Paradigm" proposal

Dear Dr. Fazal Raheman,

I am very pleased to provide a letter of support for your proposal: "ZVC (Zero Vulnerability Computing): A New Paradigm". I have a clear understanding of the above mentioned proposal conceived by Blockchain 5.0 OU, the Estonian SME that you are directing. I strongly support the development of this radical approach to proactively counter cyberattacks and I am willing to actively participate in future endeavors to bring this technology to marketplace. Any support in building this technology will positively contribute to Europe's ambition of leadership in cybersecurity.

SecLab (<https://seclab.cs.unipi.gr/>), the security research lab of the Department of Informatics at the University of Piraeus, which I am currently leading, is actively involved in various cybersecurity R&D projects, both at an EU and national level.

In light of this proposal, SecLab would be glad to provide support in the initial testing of this technology by organizing a security challenge, if possible in the context of national or international cybersecurity exercises in which we actively participate for several years.

Sincerely,

Dr. Panayiotis Kotzanikolaou



Department of Computer Science and Telecommunications
Intelligent Systems for Orchestrating Pervasive Computing
Applications (METIS) Research Lab
Papasiopoulou 2-4, 35131 Lamia Greece

Lamia, September 30th, 2021

To whom it may concern

Letter of Support

This is to acknowledge that we have reviewed the "ZVC (Zero Vulnerability Computing): A New Paradigm" proposal and find it a radical but compelling approach to sanitizing computers against cyber threats. We will be happy to associate with this project as a third party auditor of the software developed in this project. As researchers in pervasive computing and pervasive data science, we have the expertise and wherewithal to audit the source code developed in this project, and deliver a professional audit report. If there are further questions the undersigned can be contacted at the contact details given below.

Sincerely yours,

Konstantinos Kolomvatsos PhD
Director,
Intelligent Systems for Orchestrating Pervasive Computing
Applications (METIS) Research Lab
email: kostasks@uth.gr

Market Size

(Blockchain Tech Market)

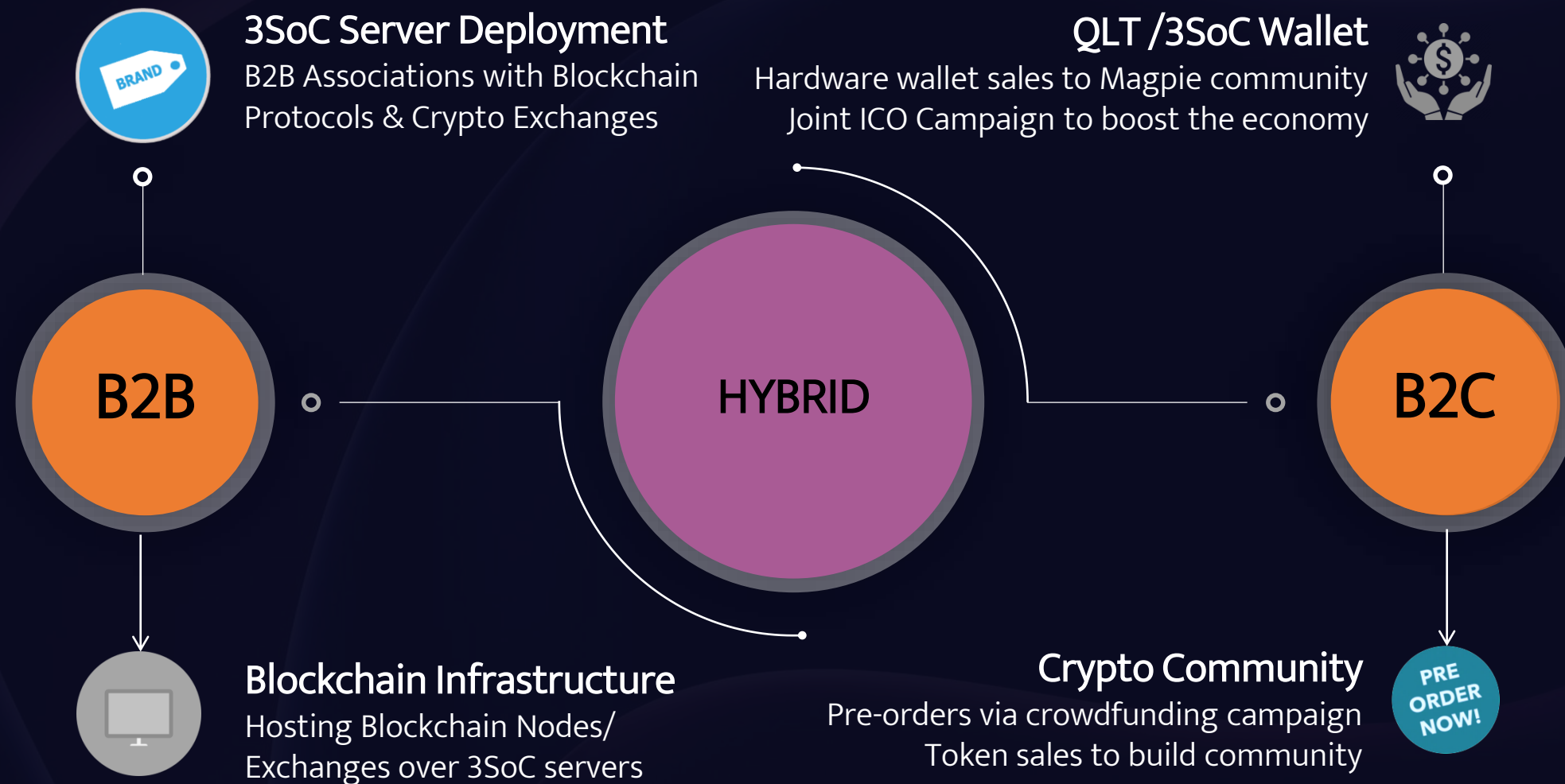
€95M net present value (NPV)
based on a conservative 1% penetration of serviceable available market (SAM) in 5 years



*Projected Year 2029 based on PrecedenceResearch.com report

Business/Revenue Model

- QLT Adopts a Unified B2B-B2C model to drive adoption across blockchain ecosystems



Value Proposition: Innovation Leadership



#1 Encryption-Agnostic Quantum-Resilient Cybersecurity

All PQC solutions claiming Q-Day security have so far failed. Even if any succeeds PQC is resource-intensive and very expensive



#2 Holistic BC/DLT Agnostic solution against all breaches

QLT can be deployed to secure any BC/DLT, NFT network or crypto exchange infrastructure against conventional as well as Q-Day threats



#3 Revenue Generating Operations: \$33M (Magpie + Fjord)

Collective cumulative revenues from Magpie Aggregator + Fjord Launchpad estimated at \$33 Million

#4 Collective User Base of ~50K users

Easy deployment with minimal changes in current infrastructure and no extravagant costings of deploying costly PQC methods, delivers great value for money

#5 QLT Super App Development for DeFi Industry

Clear Plan for QLT deployment for a super secure cross-liquidity aggregator, Magpie Protocol

Leadership Team

132+ Years of Combined Industry Experience



David Bell

CEO

Corporate strategist, crypto investor, expert in building and managing crypto networks, adroit in offering executive oversight



Fazal Raheman

Inventor, QLT

Research scientist, innovator, visionary and serial inventor with over 37 global patents.



Michael Schuette

CTO

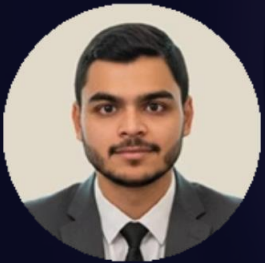
Research Scientist, Expert in NVM, cybersecurity and Solid-State drive technology with over 53 Patents in the same field



Tejas Bhagat

Head, QLT Program

Program Manager, Grant writer, architect of deep tech ecosystems, program lead for lab-to-market value creation



Ali Raheman

CSO

Cofounder of 2 successful start ups and Lead for positioning the investment strategies for QLT



Ikram Ansari

Head, Marketing

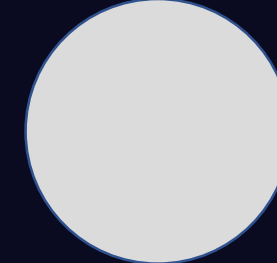
Cofounder and Dynamic professional with over 8 years of experience in sales, marketing and relationship management in B2C, B2B



Rushi Chavan

Head, Software Dev

Software Engineer with specialization in Blockchain, smart contract specialist, full-stack developer, 10+ years in decentralized systems



Name

Head, Software Dev

Software Engineer with specialization in Blockchain, smart contract specialist, full-stack developer, 10+ years in decentralized systems